NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-39003/2E
10 June 1993
SUPERSEDING
MIL-C-39003/2D
2 February 1984

PERFORMANCE SPECIFICATION SHEET

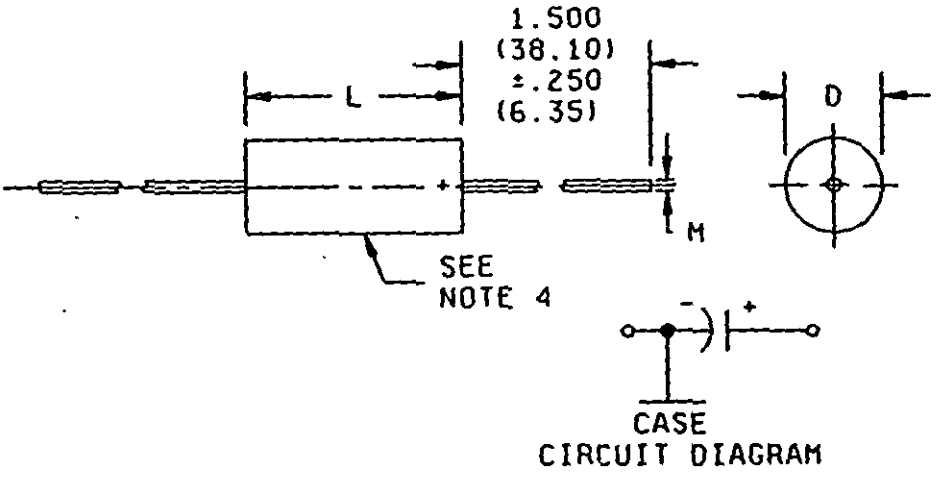
CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), ESTABLISHED RELIABILITY, STYLE CSR09

FAILURE RATE LEVELS "M", "P", AND "R" ARE INACTIVE FOR NEW DESIGN AFTER 2 FEBRUARY 1984.

FAILURE RATE LEVEL "S" IS INACTIVE FOR NEW DESIGN AFTER 15 AUGUST 1984.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39003.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. These capacitors are not intended to be mounted by their leads.
5. The case insulation shall extend .015 (0.38 mm) minimum beyond each end. However, when a shrink-fitted insulation is used, it shall lap over the ends of the capacitor body.
6. Lead length may be a minimum of 1-inch long for use in tape and reel automatic insertion equipment, when specified.

FIGURE 1. Dimensions and configuration.

(E) denotes changes

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Case size	Dimensions								
	L			D			M		
	Min	Nom	Max	Min	Nom	Max	Min	Nom	Max
A1	.235 (5.97)	.250 (6.35)	.281 (7.14)	.081 (2.06)	.090 (2.29)	.099 (2.51)	.015 (0.38)	.016 (0.41)	.021 (0.53)
B1	.375 (9.53)	.390 (9.91)	.405 (10.29)	.128 (3.25)	.138 (3.51)	.148 (3.76)	.015 (0.38)	.016 (0.41)	.021 (0.53)

FIGURE 1. Dimensions and configurations - Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tubular, insulated.

Material: Metal (alloy unspecified).

Type of seal: Hermetic.

Leads: 1/

Anode lead: Nickel-iron-alloy.

Cathode lead: Nickel-iron-alloy or Nickel.

DC rated voltage: See table I. Above 85°C voltage derating is required (see MIL-C-39003).

DC leakage (DCL): See table I.

Capacitance (Cap.)(nom): See table I.

Cap. tolerance: 5 or 10 percent, see table I.

Dissipation factor (DF): See table I.

Operating temperature: -55°C to +125°C.

Failure rate level: See table I.

Stability at low and high temperatures: In accordance with MIL-C-39003.

Step 1 (+25°C):

DCL: See table I.

Cap.: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

Cap.: Shall not change more than ±8 percent of step 1 measured value.

DF: See table I.

1/ Gold-plated leads optional, see MIL-C-39003 ordering data.

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Step 3 (+25°C):

DCL: See table I.

Cap.: Shall not change more than ±2 percent of step 1 measured value.

DF: See table I.

Step 4 (+85°C):

DCL: See table I.

Cap.: Shall not change more than ±8 percent of step 1 measured value (6, 10, 15 volt units).

Cap.: Shall not change more than ±6 percent of step 1 measured value (20 volts and higher).

DF: See table I.

Step 5 (+125°C):

DCL: See table I.

Cap.: Shall not change more than ±12 percent of step 1 measured value.

DF: See table I.

Step 6 (+25°C):

DCL: See table I.

Cap.: Shall not change more than ±2 percent of step 1 measured value.

DF: See table I.

Surge voltage: In accordance with MIL-C-39003.

DCL: See table I.

Cap.: Shall not change more than ±2 percent from initial measured value.

DF: See table I.

Life: Method 108 of MIL-STD-202.

240, 1,000, and 2,000 hours:

At 25°C (as applicable):

DCL: See table I.

(E) Cap.: In accordance with MIL-C-39003.

DF: See table I.

At 85°C:

DCL: See table I.

At 125°C:

DCL: See table I.

Marking: In accordance with MIL-C-39003.

Part or Identifying Number (PIN): M39003/02-(dash number from table I).

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(E) TABLE I. Style CSR09 characteristics and dash numbers.

DC rated voltage	Capacitance (nom)	Capacitance tolerance	DC leakage at			Dissipation factor at			Case size	Dash no. Failure rate level for (%1,000 hrs)						
			+25°C	+85°C	+125°C	-55°C	+25°C	+85°C +125°C		M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
Volts	µF	Percent	µA	µA	µA	Percent	Percent	Percent								
6	2.7	5	0.6	6.0	8.0	8	6	6	A1	1001	1061	1121	1181	2001	3001	4001
6	2.7	10	0.6	6.0	8.0	8	6	6	A1	0001	0061	0121	0181	2002	3002	4002
6	18.0	5	1.4	21.6	36.7	8	6	6	B1	1002	1062	1122	1182	2003	3003	4003
6	18.0	10	1.4	21.6	36.7	8	6	6	B1	0002	0062	0122	0182	2004	3004	4004
10	1.8	5	0.6	6.0	8.0	8	6	6	A1	1007	1067	1127	1187	2005	3005	4005
10	1.8	10	0.6	6.0	8.0	8	6	6	A1	0007	0067	0127	0187	2006	3006	4006
10	2.2	5	0.6	6.0	8.0	8	6	6	A1	1008	1068	1128	1188	2007	3007	4007
10	2.2	10	0.6	6.0	8.0	8	6	6	A1	0008	0068	0128	0188	2008	3008	4008
10	10.0	5	2.5	20.0	34.0	8	6	6	B1	1009	1069	1129	1189	2009	3009	4009
10	10.0	10	2.5	20.0	34.0	8	6	6	B1	0009	0069	0129	0189	2010	3010	4010
10	12.0	5	2.5	24.0	40.8	8	6	6	B1	1010	1070	1130	1190	2011	3011	4011
10	12.0	10	2.5	24.0	40.8	8	6	6	B1	0010	0070	0130	0190	2012	3012	4012
10	15.0	5	2.5	30.0	51.0	8	6	6	B1	1011	1071	1131	1191	2013	3013	4013
10	15.0	10	2.5	30.0	51.0	8	6	6	B1	0011	0071	0131	0191	2014	3014	4014
15	1.0	5	0.6	6.0	8.0	8	6	6	A1	1012	1072	1132	1192	2015	3015	4015
15	1.0	10	0.6	6.0	8.0	8	6	6	A1	0012	0072	0132	0192	2016	3016	4016
15	1.2	5	0.6	6.0	8.0	8	6	6	A1	1013	1073	1133	1193	2017	3017	4017
15	1.2	10	0.6	6.0	8.0	8	6	6	A1	0013	0073	0133	0193	2018	3018	4018
15	1.5	5	0.6	6.0	8.0	8	6	6	A1	1014	1074	1134	1194	2019	3019	4019
15	1.5	10	0.6	6.0	8.0	8	6	6	A1	0014	0074	0134	0194	2020	3020	4020
15	8.2	5	1.8	24.6	42.0	8	6	6	B1	1015	1075	1135	1195	2021	3021	4021
15	8.2	10	1.8	24.6	42.0	8	6	6	B1	0015	0075	0135	0195	2022	3022	4022
20	.56	5	0.6	6.0	8.0	4	3	4	A1	1016	1076	1136	1196	2023	3023	4023
20	.56	10	0.6	6.0	8.0	↓	↓	↓	A1	0016	0076	0136	0196	2024	3024	4024
20	.68	5	0.6	6.0	8.0	↓	↓	↓	A1	1017	1077	1137	1197	2025	3025	4025
20	.68	10	0.6	6.0	8.0	↓	↓	↓	A1	0017	0077	0137	0197	2026	3026	4026
20	.82	5	0.6	6.0	8.0	↓	↓	↓	A1	1018	1078	1138	1198	2027	3027	4027
20	.82	10	0.6	6.0	8.0	↓	↓	↓	A1	0018	0078	0138	0198	2028	3028	4028
20	1.0	5	0.6	6.0	8.0	↓	↓	↓	A1	1019	1079	1139	1199	2029	3029	4029
20	1.0	10	0.6	6.0	8.0	↓	↓	↓	A1	0019	0079	0139	0199	2030	3030	4030
20	3.3	5	1.0	13.2	22.4	↓	↓	↓	B1	1020	1080	1140	1200	2031	3031	4031
20	3.3	10	1.0	13.2	22.4	↓	↓	↓	B1	0020	0080	0140	0200	2032	3032	4032
20	3.9	5	2.0	15.6	26.5	↓	↓	↓	B1	1021	1081	1141	1201	2033	3033	4033
20	3.9	10	2.0	15.6	26.5	↓	↓	↓	B1	0021	0081	0141	0201	2034	3034	4034
20	4.7	5	2.0	18.8	32.0	↓	↓	↓	B1	1022	1082	1142	1202	2035	3035	4035
20	4.7	10	2.0	18.8	32.0	↓	↓	↓	B1	0022	0082	0142	0202	2036	3036	4036
20	5.6	5	2.0	22.4	38.1	↓	↓	↓	B1	1023	1083	1143	1203	2037	3037	4037
20	5.6	10	2.0	22.4	38.1	↓	↓	↓	B1	0023	0083	0143	0203	2038	3038	4038
20	6.8	5	2.0	27.2	46.2	↓	↓	↓	B1	1024	1084	1144	1204	2039	3039	4039
20	6.8	10	2.0	27.2	46.2	↓	↓	↓	B1	0024	0084	0144	0204	2040	3040	4040
35	0.33	5	0.6	6.0	8.0	↓	↓	↓	A1	1025	1085	1145	1205	2041	3041	4041
35	0.33	10	0.6	6.0	8.0	↓	↓	↓	A1	0025	0085	0145	0205	2042	3042	4042
35	0.39	5	0.6	6.0	8.0	↓	↓	↓	A1	1026	1086	1146	1206	2043	3043	4043
35	0.39	10	0.6	6.0	8.0	↓	↓	↓	A1	0026	0086	0146	0206	2044	3044	4044
35	0.47	5	0.6	6.0	8.0	↓	↓	↓	A1	1027	1087	1147	1207	2045	3045	4045
35	0.47	10	0.6	6.0	8.0	↓	↓	↓	A1	0027	0087	0147	0207	2046	3046	4046
35	2.2	5	1.4	15.4	26.2	↓	↓	↓	B1	1028	1088	1148	1208	2047	3047	4047
35	2.2	10	1.4	15.4	26.2	↓	↓	↓	B1	0028	0088	0148	0208	2048	3048	4048
35	2.7	5	1.4	18.9	32.1	↓	↓	↓	B1	1029	1089	1149	1209	2049	3049	4049
35	2.7	10	1.4	18.9	32.1	↓	↓	↓	B1	0029	0089	0149	0209	2050	3050	4050

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(E) TABLE I. Style CSR09 characteristics and dash numbers - Continued.

DC rated voltage	Capacitance (nom)	Capacitance tolerance	DC leakage at			Dissipation factor at			Case size	Dash no. Failure rate level for (%1,000 hrs)						
			+25°C	+85°C	+125°C	-55°C	+25°C	+85°C +125°C		M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
Volts	µF	Percent	µA	µA	µA	Percent	Percent	Percent								
50	0.22	5	0.6	6.0	8.0	4	3	4	A1	1030	1090	1150	1210	2051	3051	4051
50	0.22	10	0.6	6.0	8.0				A1	0030	0090	0150	0210	2052	3052	4052
50	0.27	5	0.6	6.0	8.0				A1	1031	1091	1151	1211	2053	3053	4053
50	0.27	10	0.6	6.0	8.0				A1	0031	0091	0151	0211	2054	3054	4054
50	1.5	5	1.4	15.0	25.5				B1	1032	1092	1152	1212	2055	3055	4055
50	1.5	10	1.4	15.0	25.5				B1	0032	0092	0152	0212	2056	3056	4056
50	1.8	5	1.4	18.0	30.6				B1	1033	1093	1153	1213	2057	3057	4057
50	1.8	10	1.4	18.0	30.6				B1	0033	0093	0153	0213	2058	3058	4058
75	0.047	5	0.6	6.0	8.0				A1	1034	1094	1154	1214	2059	3059	4059
75	0.047	10	0.6	6.0	8.0				A1	0034	0094	0154	0214	2060	3060	4060
75	0.056	5	0.6	6.0	8.0				A1	1035	1095	1155	1215	2061	3061	4061
75	0.056	10	0.6	6.0	8.0				A1	0035	0095	0155	0215	2062	3062	4062
75	0.068	5	0.6	6.0	8.0				A1	1036	1096	1156	1216	2063	3063	4063
75	0.068	10	0.6	6.0	8.0				A1	0036	0096	0156	0216	2064	3064	4064
75	0.082	5	0.6	6.0	8.0				A1	1037	1097	1157	1217	2065	3065	4065
75	0.082	10	0.6	6.0	8.0				A1	0037	0097	0157	0217	2066	3066	4066
75	0.10	5	0.6	6.0	8.0				A1	1038	1098	1158	1218	2067	3067	4067
75	0.10	10	0.6	6.0	8.0				A1	0038	0098	0158	0218	2068	3068	4068
75	0.12	5	0.6	6.0	8.0				A1	1039	1099	1159	1219	2069	3069	4069
75	0.12	10	0.6	6.0	8.0				A1	0039	0099	0159	0219	2070	3070	4070
75	0.15	5	0.6	6.0	8.0				A1	1040	1100	1160	1220	2071	3071	4071
75	0.15	10	0.6	6.0	8.0				A1	0040	0100	0160	0220	2072	3072	4072
75	0.18	5	0.6	6.0	8.0				A1	1041	1101	1161	1221	2073	3073	4073
75	0.18	10	0.6	6.0	8.0				A1	0041	0101	0161	0221	2074	3074	4074
75	0.22	5	0.6	6.0	8.0				B1	1042	1102	1162	1222	2075	3075	4075
75	0.22	10	0.6	6.0	8.0				B1	0042	0102	0162	0222	2076	3076	4076
75	0.27	5	0.6	6.0	8.0				B1	1043	1103	1163	1223	2077	3077	4077
75	0.27	10	0.6	6.0	8.0				B1	0043	0103	0163	0223	2078	3078	4078
75	0.33	5	0.6	6.0	8.4				B1	1044	1104	1164	1224	2079	3079	4079
75	0.33	10	0.6	6.0	8.4				B1	0044	0104	0164	0224	2080	3080	4080
75	0.39	5	0.6	6.0	9.9				B1	1045	1105	1165	1225	2081	3081	4081
75	0.39	10	0.6	6.0	9.9				B1	0045	0105	0165	0225	2082	3082	4082
75	0.47	5	0.6	7.0	12.0				B1	1046	1106	1166	1226	2083	3083	4083
75	0.47	10	0.6	7.0	12.0				B1	0046	0106	0166	0226	2084	3084	4084
75	0.56	5	0.6	8.4	14.3				B1	1047	1107	1167	1227	2085	3085	4085
75	0.56	10	0.6	8.4	14.3				B1	0047	0107	0167	0227	2086	3086	4086
75	0.68	5	0.6	10.2	17.3				B1	1048	1108	1168	1228	2087	3087	4087
75	0.68	10	0.6	10.2	17.3				B1	0048	0108	0168	0228	2088	3088	4088
75	0.82	5	0.7	12.3	20.9				B1	1049	1109	1169	1229	2089	3089	4089
75	0.82	10	0.7	12.3	20.9				B1	0049	0109	0169	0229	2090	3090	4090
75	1.0	5	0.9	15.0	25.5				B1	1050	1110	1170	1230	2091	3091	4091
75	1.0	10	0.9	15.0	25.5				B1	0050	0110	0170	0230	2092	3092	4092
75	1.2	5	0.9	18.0	30.6				B1	1051	1111	1171	1231	2093	3093	4093
75	1.2	10	0.9	18.0	30.6				B1	0051	0111	0171	0231	2094	3094	4094

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Substitutability data:

Substitutability for failure rate levels shall be as follows: In accordance with MIL-C-39003.

Substitutability for capacitance tolerance shall be as follows:

Cap tolerance	Will replace cap tolerance
5%	10%

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Navy - SH
Air Force - 17, 85, 99
DLA - ES

User activities:
Navy - MC, AS, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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